MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

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MOTIVATE INNOVATE EMPOWER 29 YEARS

Unit: II

Industrial Waste Water Management

Industrial waste water Management

- Involves the collection, treatment, reuse, and safe disposal of wastewater generated from industries.
- High organic load, toxic chemicals, heavy metals, oils, acids, alkalis, and suspended solids, which can seriously affect water bodies, soil, and public health if discharged untreated.

Objectives

- To reduce pollution load before discharge.
- To protect receiving water bodies.
- To comply with effluent standards.
- To reuse and recycle water.
- To ensure environmental sustainability.

Strategies for Pollution Control

- **Volume Reduction**
- **Strength Reduction**

Volume Reduction

Methods:

- Process modification
- Water conservation
- Reuse and recycling
- Segregation of wastewater

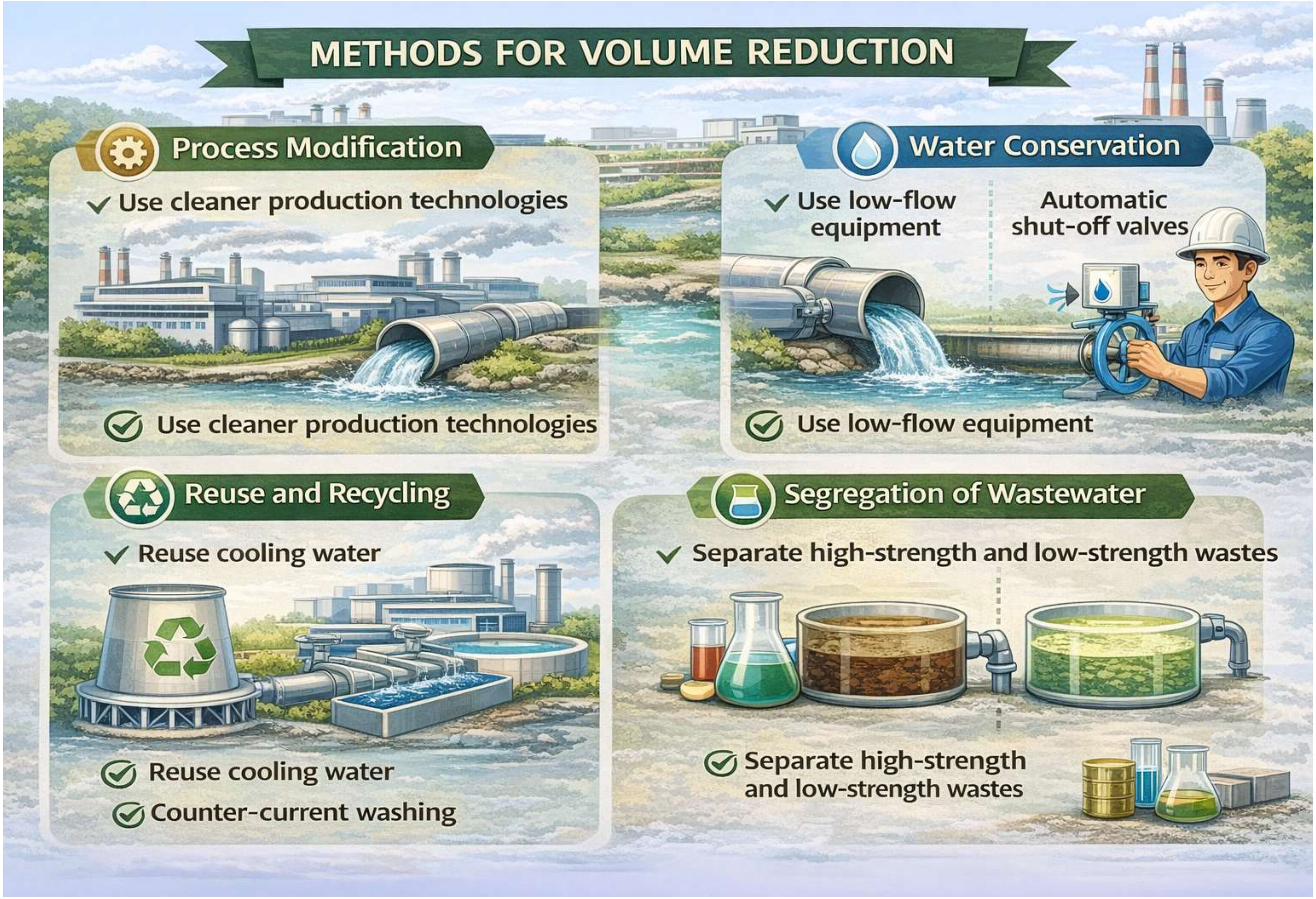
Strength Reduction

Methods:

- Raw material substitution
- Process control
- Good housekeeping
- By-product recovery

Volume Reduction

- Volume reduction of industrial wastewater refers to the systematic minimization of the quantity of wastewater generated, handled, treated, or disposed of in an industrial facility, without compromising production quality.
- Volume reduction is a primary pollution-prevention strategy in industrial wastewater management, emphasizing source control rather than end-of-pipe treatment



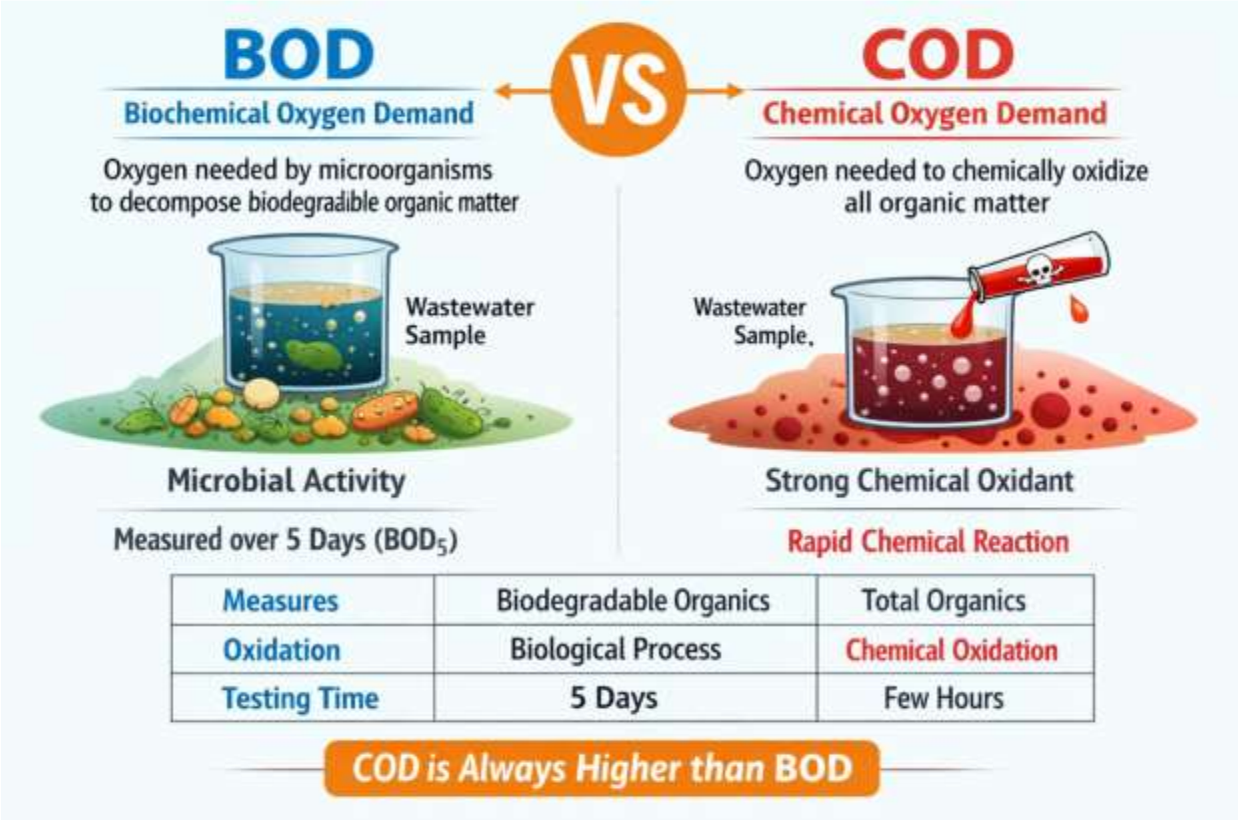
METHODS FOR VOLUME REDUCTION
Process Modification
Use cleaner production technologies
Use cleaner production technologies
Water Conservation
Use low-flow equipment
Automatic shut-off valves
Use low-flow equipment
Reuse and Recycling
Reuse cooling water
Reuse cooling water
Counter-current washing
Segregation of Wastewater
Separate high-strength and low-strength wastes
Separate high-strength and low-strength wastes

- **Process modification** involves redesigning or optimizing industrial operations to reduce water demand. This includes adopting dry or semi-dry processes, improving equipment efficiency, and implementing closed-loop systems that limit wastewater generation.
- **Water conservation** aims at reducing unnecessary water use through good housekeeping practices, regular maintenance to prevent leaks, use of efficient fixtures, and monitoring of water consumption within the plant.
- **Reuse and recycling** of wastewater involves treating effluent to a suitable level and reusing it within the industry for applications such as cooling, washing, or auxiliary processes, thereby reducing fresh water intake and wastewater discharge.
- **Segregation of wastewater** refers to separating different wastewater streams according to their contamination level. Highly polluted streams are treated separately, while low-strength or relatively clean streams can be reused or require minimal treatment, reducing the overall volume to be treated.


ADVANTAGES OF VOLUME REDUCTION
Reduces Treatment Cost
Less wastewater means lower treatment expenses
Less wastewater means lower treatment expenses
Improves Efficiency
Smaller, more effective treatment units
HIGH
Smaller, more effective treatment units
Conserves Freshwater Resources
Reduces demand on natural water sources
Reduces demand on natural water sources

Strength Reduction

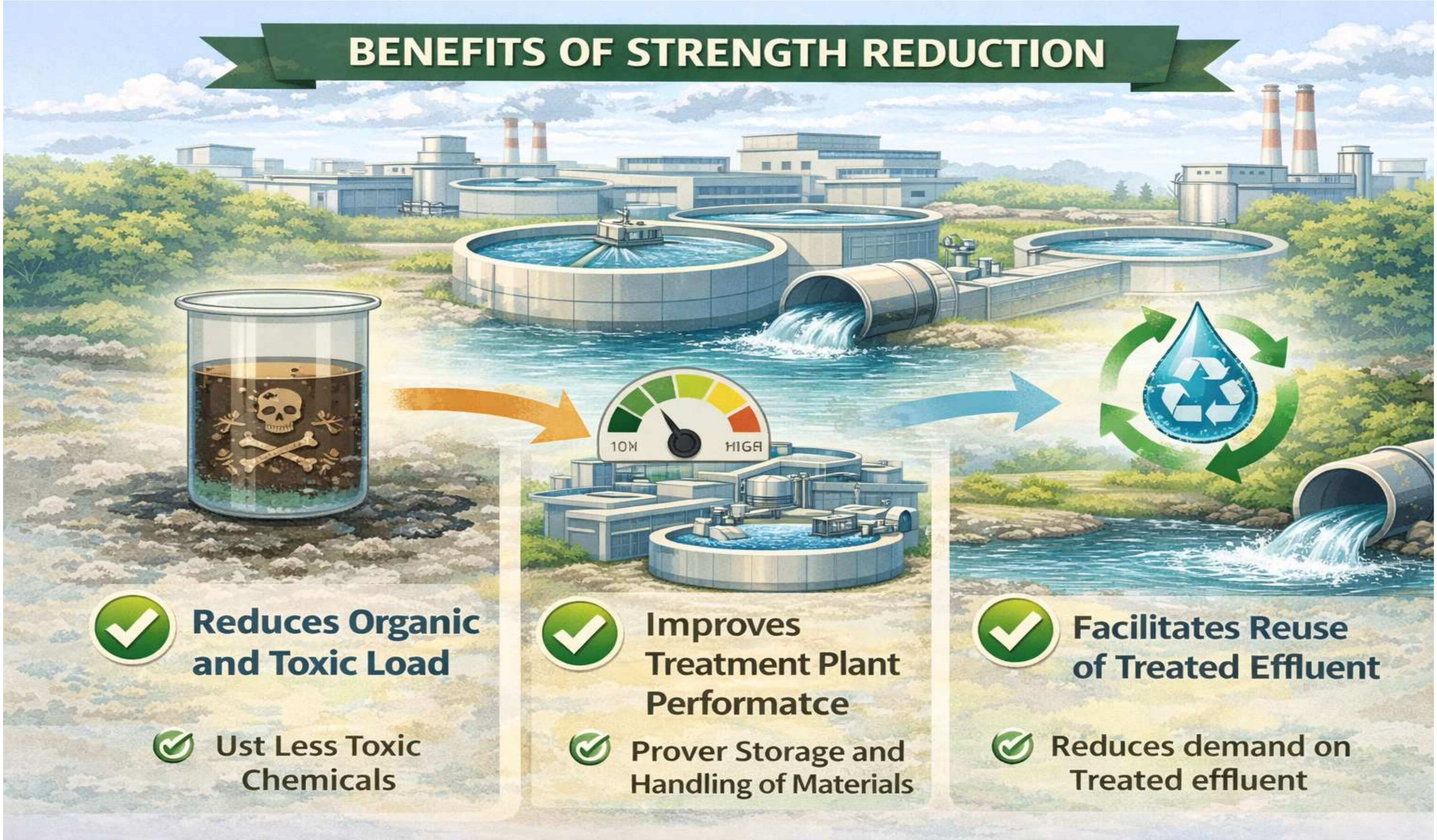
- **Strength reduction of industrial wastewater** refers to the **reduction of pollutant concentration** (such as BOD, COD, TSS, oil and grease, toxic substances) before discharge or biological treatment. Unlike volume reduction, it focuses on **lowering contaminant load** rather than flow rate.



Measures	Biodegradable Organics	Total Organics
Oxidation	Biological Process	Chemical Oxidation
Testing Time	5 Days	Few Hours


METHODS FOR STRENGTH REDUCTION
Raw Material Substitution
Less Toxic
Use Less Toxic Chemicals
Prevent Spills and Leakages
Process Control
Prevent Spills and Leakages
Good Housekeeping
Proper Storage and Handling of Materials
By-Product Recovery
Recovery of Acids, Alkalis, Solvents, Metals

- **Raw material substitution** involves replacing toxic, high-strength, or non-biodegradable raw materials with less polluting and environmentally safer alternatives. This reduces the concentration of harmful contaminants in the wastewater.
- **Process control** refers to proper monitoring and regulation of industrial operations such as flow rate, temperature, reaction time, and chemical dosage. Effective process control minimizes incomplete reactions, chemical losses, and accidental discharges, thereby reducing wastewater strength.
- **Good housekeeping** includes operational practices such as spill prevention, proper storage of chemicals, regular equipment maintenance, and employee training. These measures prevent unnecessary contamination of wastewater and reduce pollutant load at the source.
- **By-product recovery** focuses on recovering useful materials such as chemicals, solvents, oils, or metals from waste streams. This not only reduces the strength of wastewater but also improves resource efficiency and economic viability.



BENEFITS OF STRENGTH REDUCTION
10M
HIGA
Reduces Organic and Toxic Load
Ust Less Toxic Chemicals
Improves Treatment Plant Performatce
Prover Storage and Handling of Materials
Facilitates Reuse of Treated Effluent
Reduces demand on Treated effluent

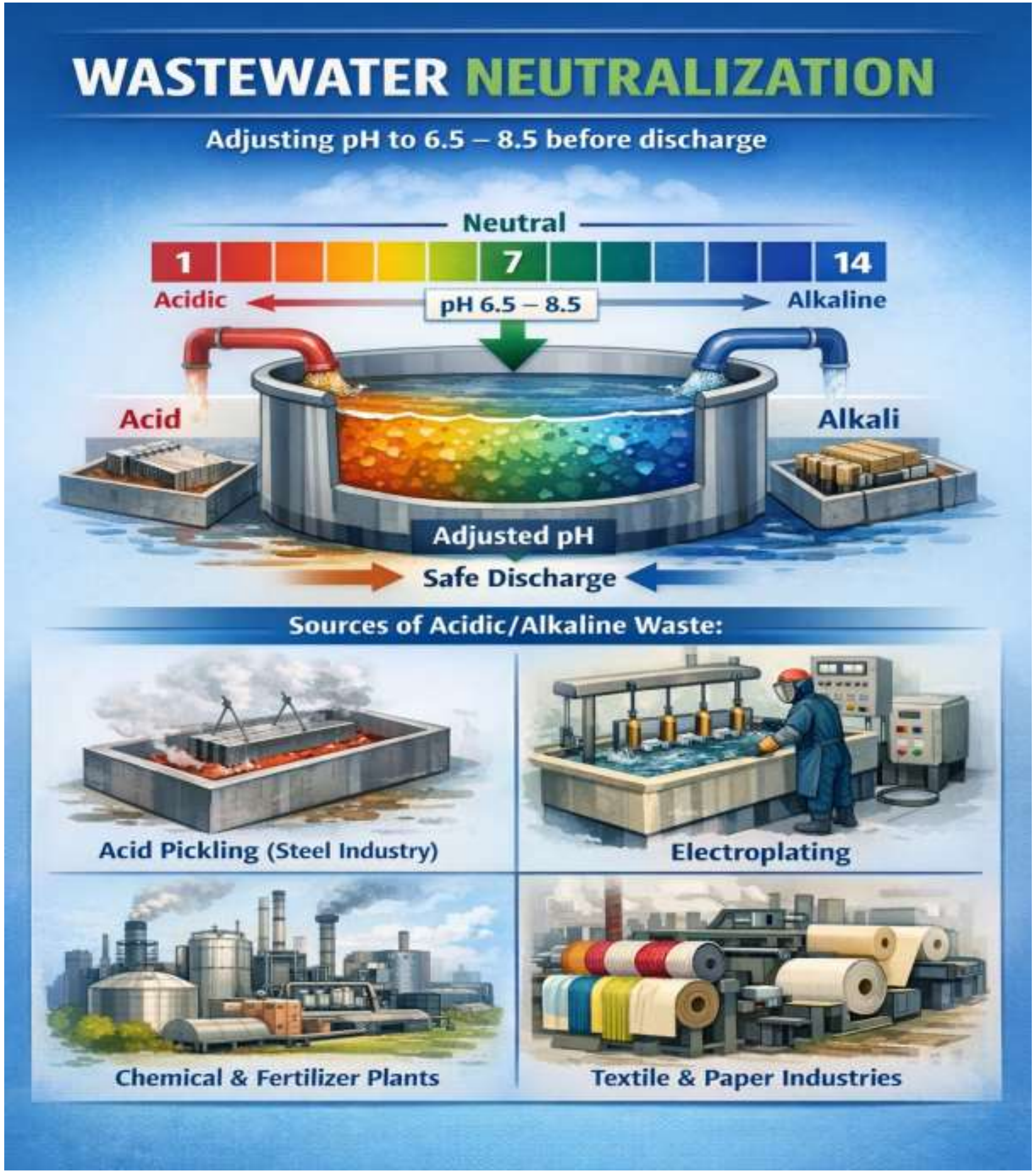
Neutralization

Definition:

Neutralization is the process of **adjusting the pH of wastewater** to an acceptable range (usually **pH 6.5 – 8.5**) before discharge or further treatment.

Sources of acidic/alkaline waste:

- Acid pickling (steel industry).
- Electroplating.
- Chemical and fertilizer industries.
- Textile and paper industries.



Methods of Neutralization

1. Chemical Neutralization

- Addition of neutralizing agents to adjust pH.
  - **For acidic wastewater:**
    - Lime ($Ca(OH)_2$)
    - Limestone ($CaCO_3$)
    - Sodium hydroxide (NaOH)
  - **For alkaline wastewater:**
    - Sulfuric acid (H_2SO_4)
    - Hydrochloric acid (HCl)
    - Carbon dioxide (CO_2)

2. Self-neutralization

- Mixing acidic and alkaline wastes

Chemical Neutralization

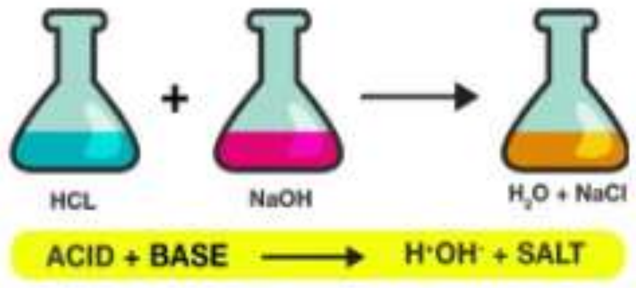
- Chemical neutralization is a widely used wastewater treatment process in which acidic or alkaline wastewater is treated by adding suitable chemicals to adjust its pH to a permissible neutral range (generally 6.5–8.5) before discharge or further treatment.

Principle

- Chemical neutralization is based on the acid–base reaction, where:
- Acids release hydrogen ions (H^+)
- Bases (alkalis) release hydroxyl ions (OH^-)
- When combined in appropriate proportions:

$$H^+ + OH^- \rightarrow H_2O$$

- This reaction forms water and salts, thereby neutralizing the wastewater.



1. Neutralization of Acidic Wastewater

- Acidic effluents ($pH < 6.5$) are treated using alkaline chemicals such as:
  - Lime ($Ca(OH)_2$)
  - Limestone ($CaCO_3$)
  - Sodium hydroxide ($NaOH$)
  - Sodium carbonate (Na_2CO_3)
- Example reaction:

$$H_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2H_2O$$

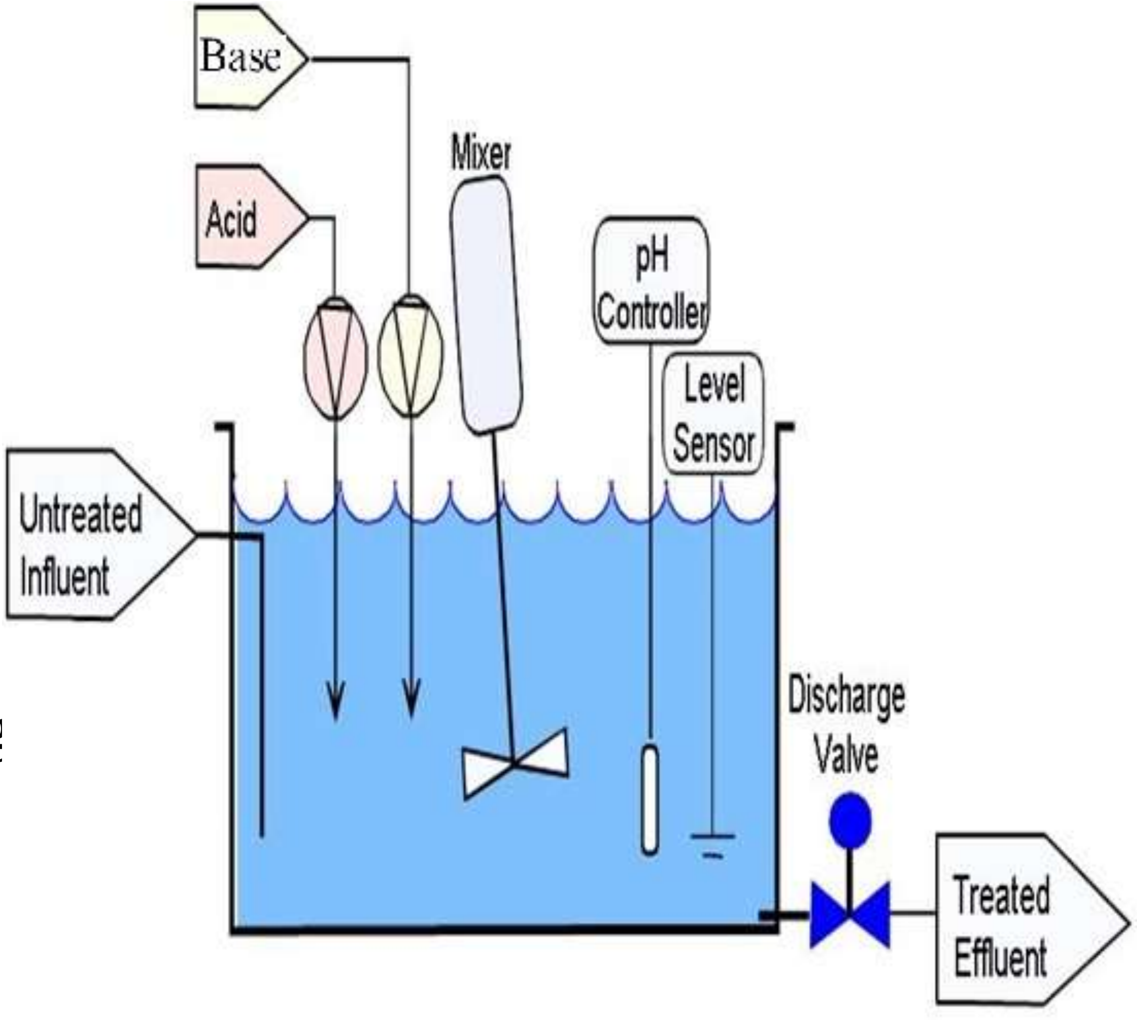
2. Neutralization of Alkaline Wastewater

- Alkaline effluents ($pH > 8.5$) are treated using acidic chemicals such as:
  - Sulfuric acid (H_2SO_4)
  - Hydrochloric acid (HCl)
  - Carbon dioxide (CO_2)
  - Nitric acid (HNO_3)
- Example reaction:

$$NaOH + HCl \rightarrow NaCl + H_2O$$

Process Description

- **Wastewater collection**
  Effluent is directed to a neutralization tank.
- **Chemical dosing**
  Required neutralizing agent is added based on influent pH.
- **Mixing**
  Mechanical or hydraulic mixers ensure uniform reaction.
- **pH monitoring and control**
  Online pH sensors regulate chemical dosing automatically.
- **Discharge or further treatment**
  Neutralized effluent is sent to secondary treatment or disposal.



Self-neutralization

Self-neutralization, also called blending, is a physical–chemical pH control method in which acidic and alkaline wastewater streams are mixed together so that their pH values counterbalance each other, resulting in wastewater with a near-neutral pH, without (or with minimal) addition of external chemicals.

Principle

- Acidic wastewater (low pH) contains excess H^+ ions.
- Alkaline wastewater (high pH) contains excess OH^- ions.
- When blended in suitable proportions:

$$H^+ + OH^- \rightarrow H_2O$$

- The final pH approaches neutrality (≈ 7).

Process Description

- **Collection of wastewater streams**
  - Acidic and alkaline effluents are collected separately from different process units.
- **Controlled mixing**
  - Streams are mixed in a neutralization or equalization tank.
- **pH monitoring**
  - Continuous or intermittent pH measurement is carried out.
- **Flow adjustment**
  - Flow rates are adjusted to achieve the desired pH.
- **Discharge or further treatment**
  - Neutralized wastewater is sent for secondary treatment or disposal.

Importance:

- Protects treatment units.
- Prevents corrosion.
- Safeguards aquatic life.

Aspect	Chemical Neutralization	Self-Neutralization (Blending)
Definition	pH adjustment by adding external chemicals (acid or alkali) to wastewater.	pH adjustment by mixing acidic and alkaline wastewater streams.
Source of neutralizing agent	External chemicals such as lime, NaOH, H_2SO_4, HCl.	Wastewater itself acts as neutralizing medium.
Principle	Acid–base reaction ($H^+ + OH^- \rightarrow H_2O$).	Acid–base reaction ($H^+ + OH^- \rightarrow H_2O$).
Chemical consumption	High, depends on wastewater strength.	Low or nil, minimal chemical use.
Operational cost	Higher due to chemical purchase and handling.	Lower and more economical.
Control of pH	Precise and reliable pH control possible.	Less precise due to flow and pH variations.
Suitability	Suitable for strongly acidic and alkaline wastewater.	Suitable only when both acidic and alkaline streams are available.
Flexibility	Can treat any pH condition.	Limited flexibity.

Equalization

Definition:

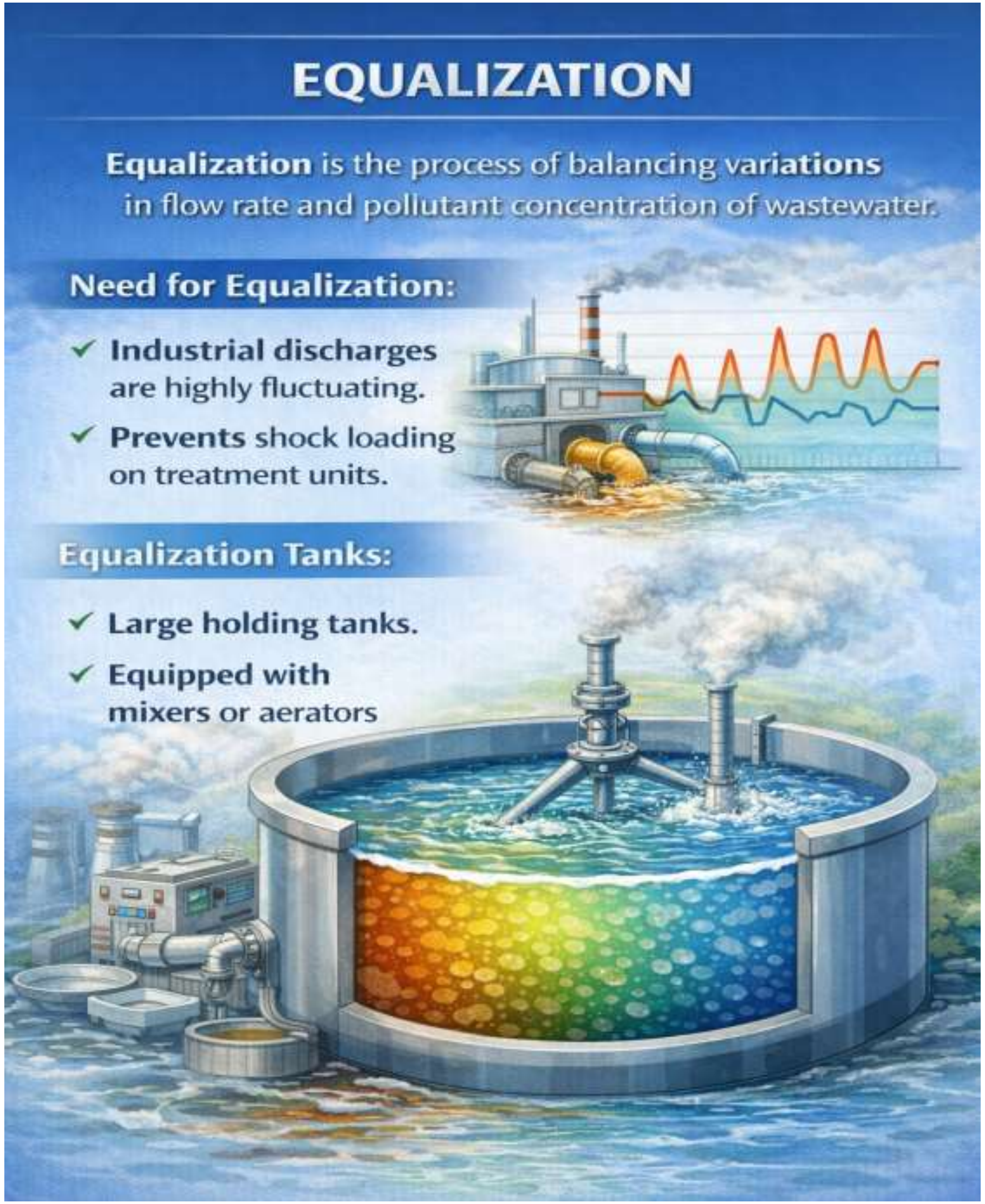
Equalization is a preliminary wastewater treatment process in which flow rate, pollutant concentration, and pH variations are minimized by temporarily storing wastewater in a dedicated equalization tank or basin.

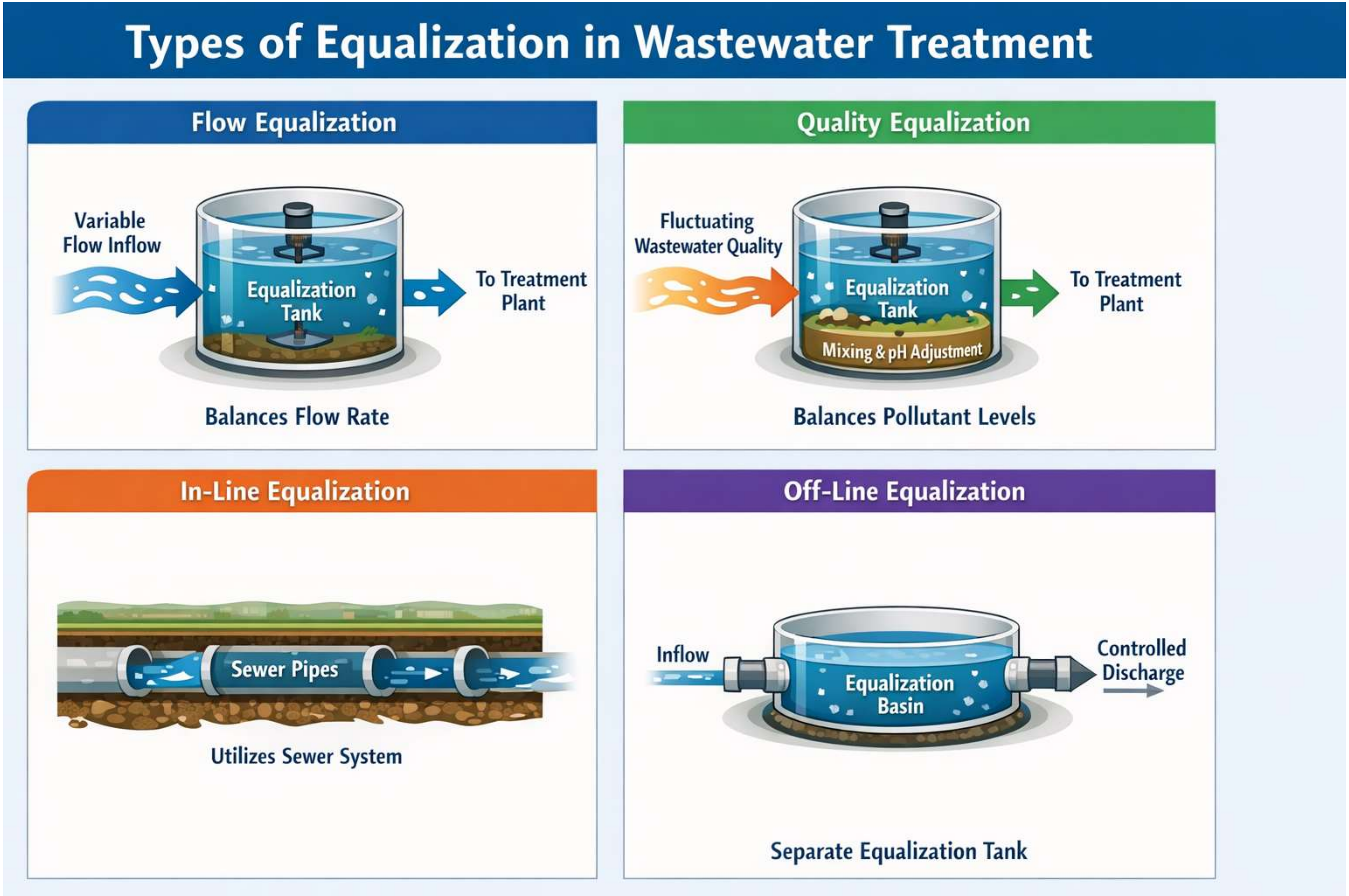
Need for Equalization:

- Industrial discharges are highly fluctuating
- Prevents shock loading on treatment units.

Equalization Tanks:

- Large holding tanks.
- Equipped with mixers or aerators.




Types of Equalization in Wastewater Treatment
Flow Equalization
Variable Flow Inflow
Equalization Tank
To Treatment Plant
Balances Flow Rate
Quality Equalization
Fluctuating Wastewater Quality
Equalization Tank
Mixing & pH Adjustment
To Treatment Plant
Balances Pollutant Levels
In-Line Equalization
Sewer Pipes
Utilizes Sewer System
Off-Line Equalization
Inflow
Equalization Basin
Controlled Discharge
Separate Equalization Tank

Types of Equalization

Equalization can be classified based on **its function and mode of operation**

1. Flow Equalization

- Reduces fluctuations in flow rate
- Maintains constant hydraulic loading

2. Quality Equalization

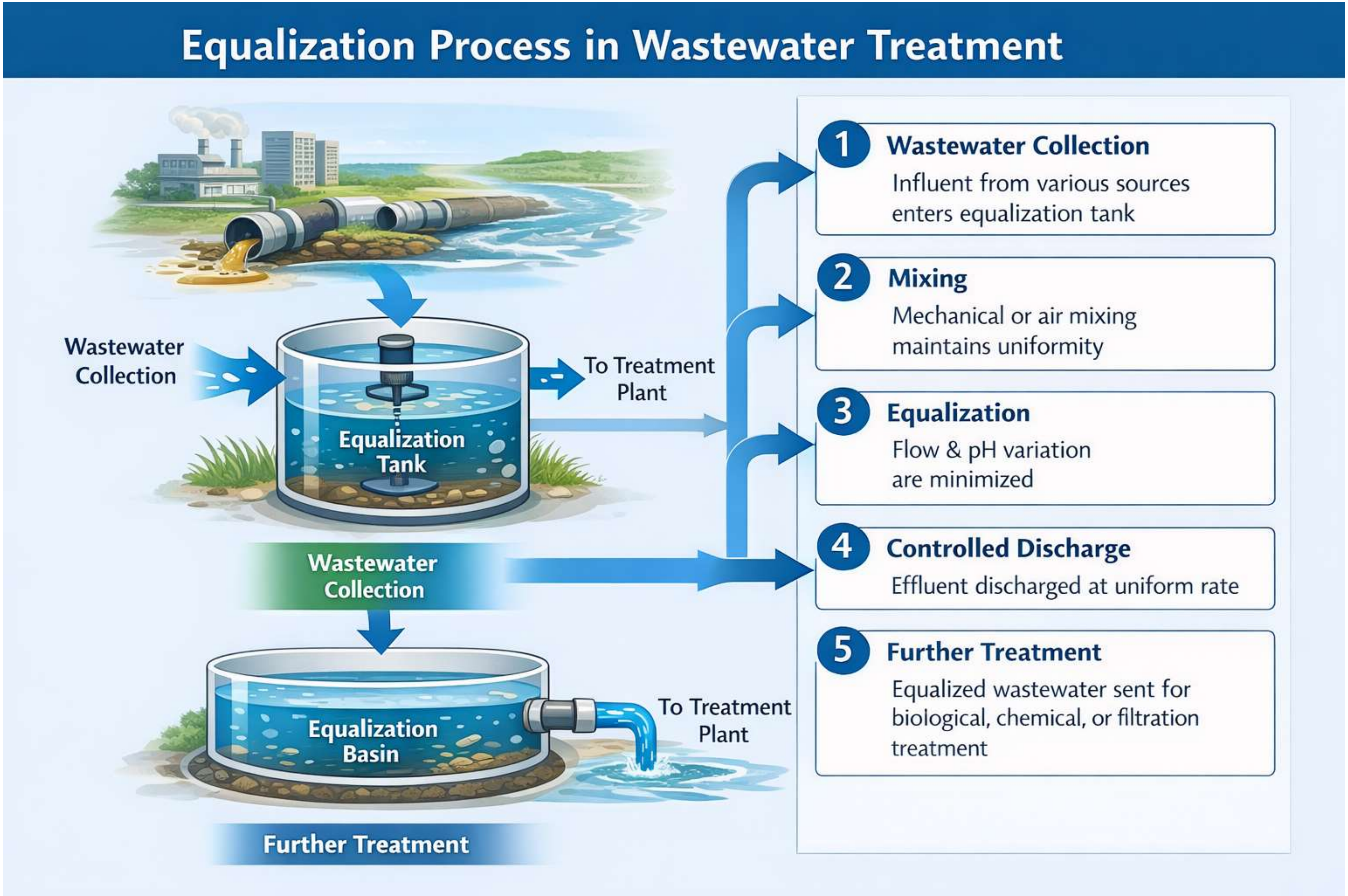
- Reduces variations in BOD, COD, TSS, pH, toxic compounds

3. In-Line Equalization

- Wastewater is stored within the sewer system
- Uses existing pipelines (limited application)

4. Off-Line Equalization

- Involves a separate equalization tank or basin where wastewater is diverted and stored.
- Achieve effective flow and quality equalization
- Most common and effective method



Equalization Process in Wastewater Treatment
Wastewater Collection
To Treatment Plant
Equalization Tank
Wastewater Collection
Equalization Basin
To Treatment Plant
Further Treatment
1 Wastewater Collection
Influent from various sources enters equalization tank
2 Mixing
Mechanical or air mixing maintains uniformity
3 Equalization
Flow & pH variation are minimized
4 Controlled Discharge
Effluent discharged at uniform rate
5 Further Treatment
Equalized wastewater sent for biological, chemical, or filtration treatment

Equalization Process

- Wastewater from various sources is collected in an **equalization tank.**
- **Continuous mixing or aeration** prevents settling and stratification.
- Wastewater is discharged at a **controlled, nearly constant rate.**
- pH and concentration become more uniform before treatment.

Functions:

- Uniform flow.
- Uniform pollutant concentration.
- Improved efficiency of biological treatment.

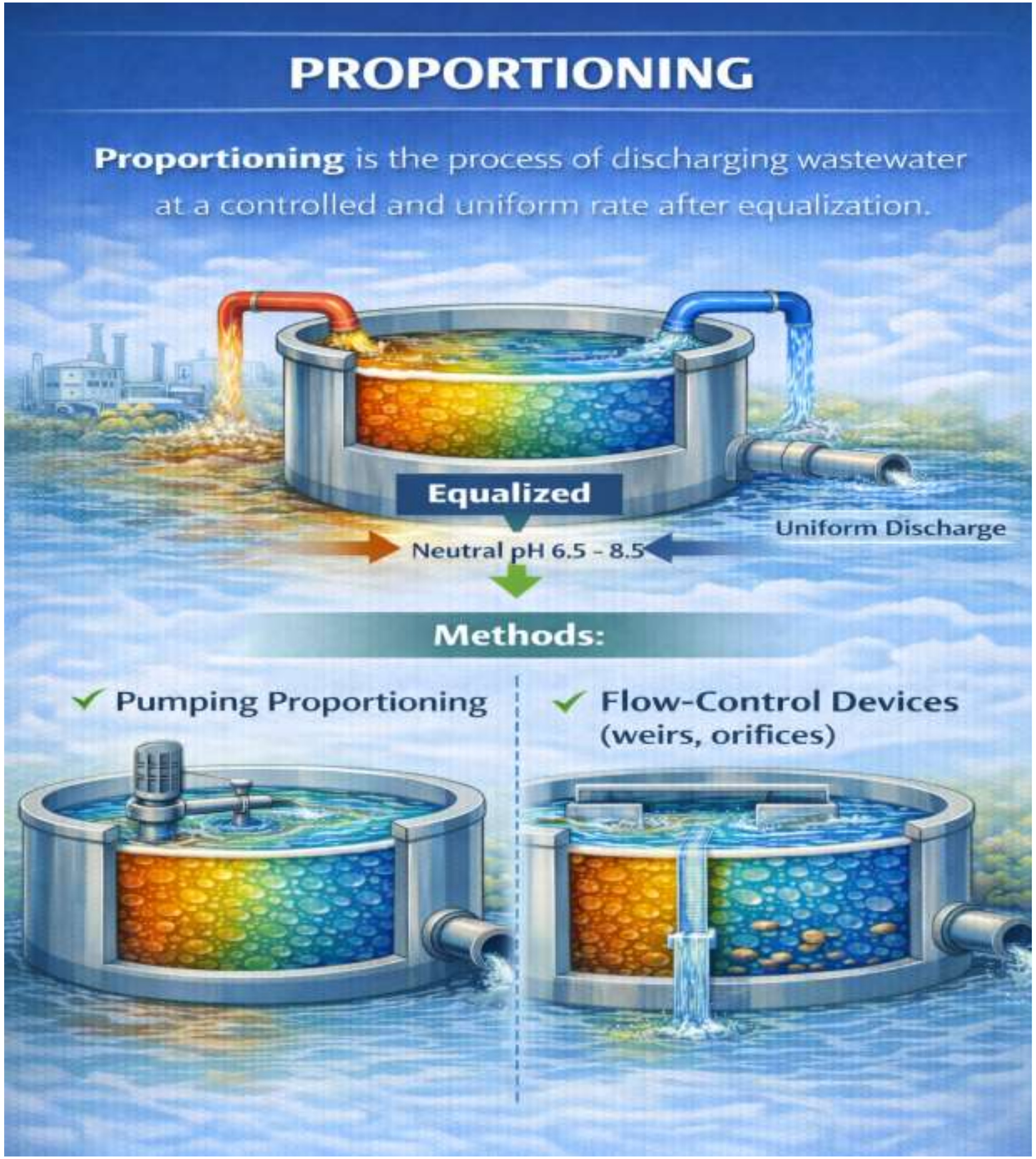
Proportioning

Definition:

- Proportioning is the process of regulating and distributing wastewater flow in a predetermined ratio or rate to ensure uniform loading on treatment units.
- In other words, Proportioning is the controlled mixing of two or more wastewater streams—usually of different flow rates or pollutant strengths—to produce a uniform combined flow.

Methods:

- Pumping proportioning.
- Flow-control devices (weirs, orifices).



Principle of Proportioning

- Wastewater stored in an equalization basin is discharged in proportion to time or treatment capacity, not inflow fluctuations.
- Flow is controlled using flow-regulating devices such as pumps, weirs, or valves.

Process Description

- Wastewater is first collected and equalized.
- Flow-measuring devices monitor discharge rate.
- Proportioning devices release wastewater at a **uniform or preset rate**.
- Downstream treatment units receive **steady inflow**.

Types of Proportioning

1. Constant-Rate Proportioning

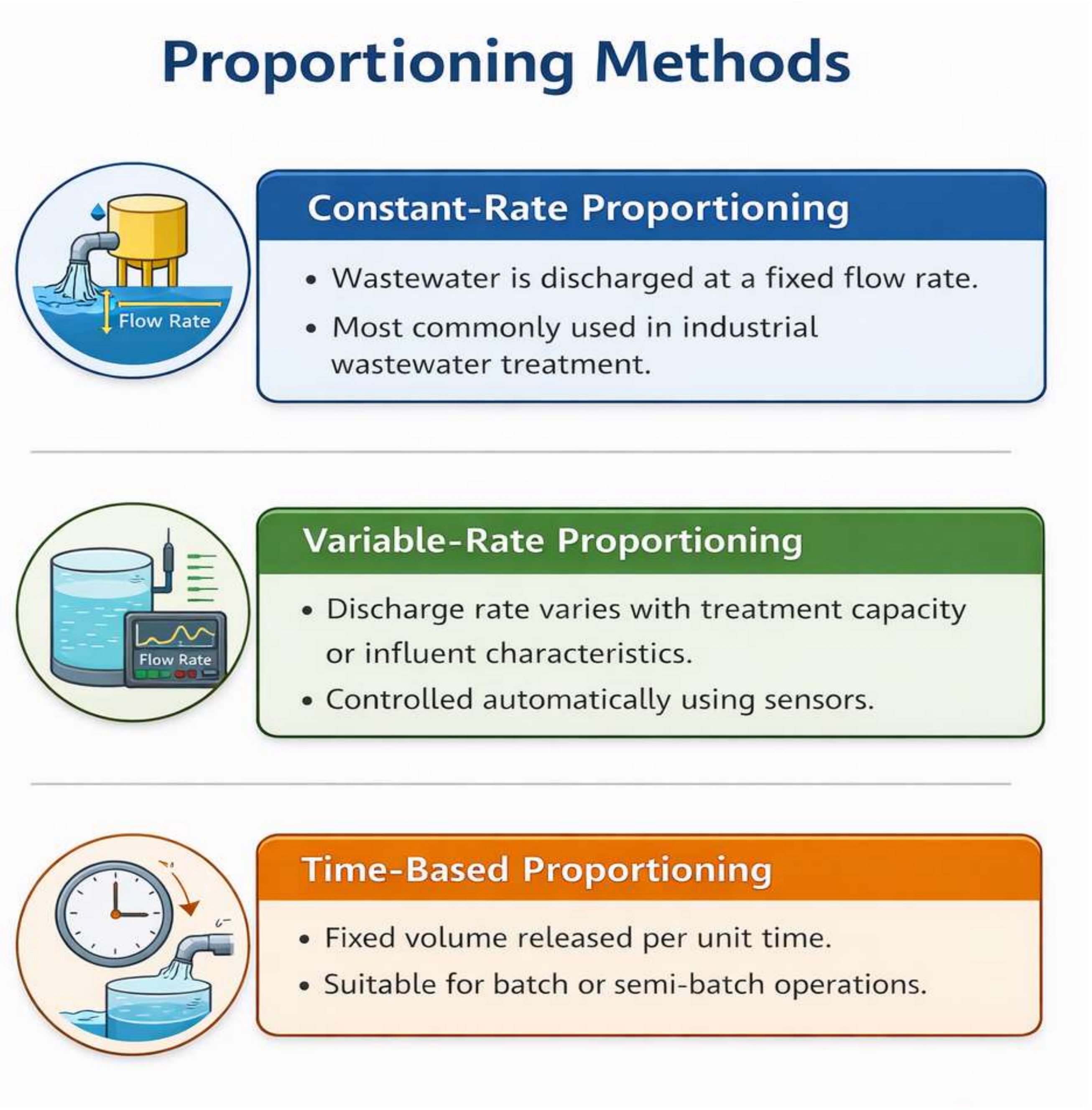
- Wastewater is discharged at a fixed flow rate.
- Most commonly used in industrial wastewater treatment.

2. Variable-Rate Proportioning

- Discharge rate varies with treatment capacity or influent characteristics.
- Controlled automatically using sensors.

3. Time-Based Proportioning

- Fixed volume released per unit time.
- Suitable for batch or semi-batch operations.



Advantages:

- Prevents overloading of treatment units.
- Improves operational stability.
- Ensures consistent treatment performance.

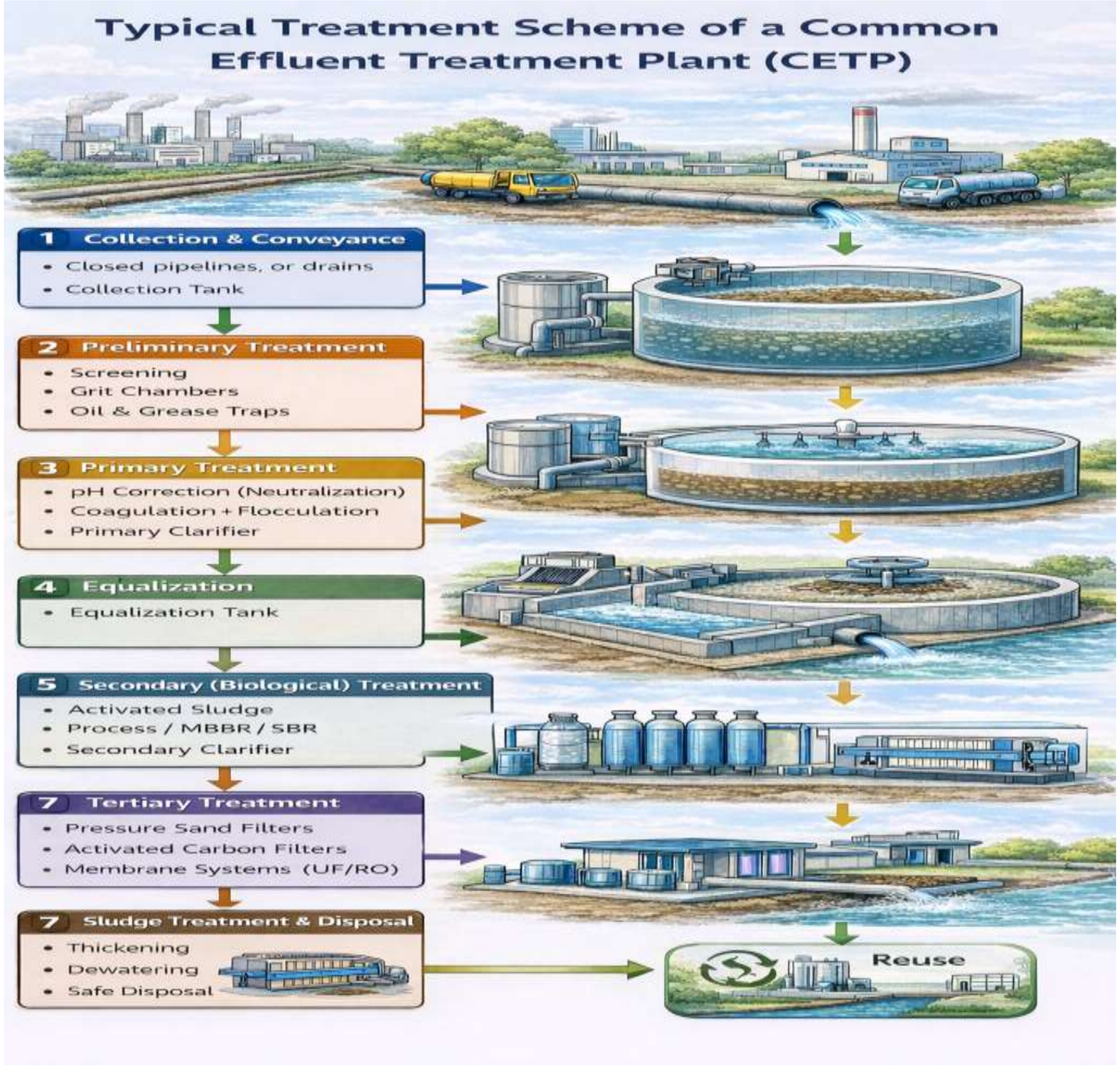
Common Effluent Treatment Plants (CETPs)

Definition:

CETP is a centralized treatment facility that treats effluents from multiple small- and medium-scale industries located in an industrial cluster.

Need for CETPs:

- Small industries cannot afford individual treatment plants.
- Economies of scale.
- Better pollution control.


Typical Treatment Scheme of a Common Effluent Treatment Plant (CETP)
1 Collection & Conveyance
• Closed pipelines, or drains
• Collection Tank
2 Preliminary Treatment
• Screening
• Grit Chambers
• Oil & Grease Traps
3 Primary Treatment
• pH Correction (Neutralization)
• Coagulation + Flocculation
• Primary Clarifier
4 Equalization
• Equalization Tank
5 Secondary (Biological) Treatment
• Activated Sludge
• Process / MBBR / SBR
• Secondary Clarifier
7 Tertiary Treatment
• Pressure Sand Filters
• Activated Carbon Filters
• Membrane Systems (UF/RO)
7 Sludge Treatment & Disposal
• Thickening
• Dewatering
• Safe Disposal
Reuse

1. Collection & Conveyance

- Effluents from different industries are collected through a common sewer network
- Flow may vary in quantity and quality depending on industrial activity

Purpose: Centralized handling of wastewater

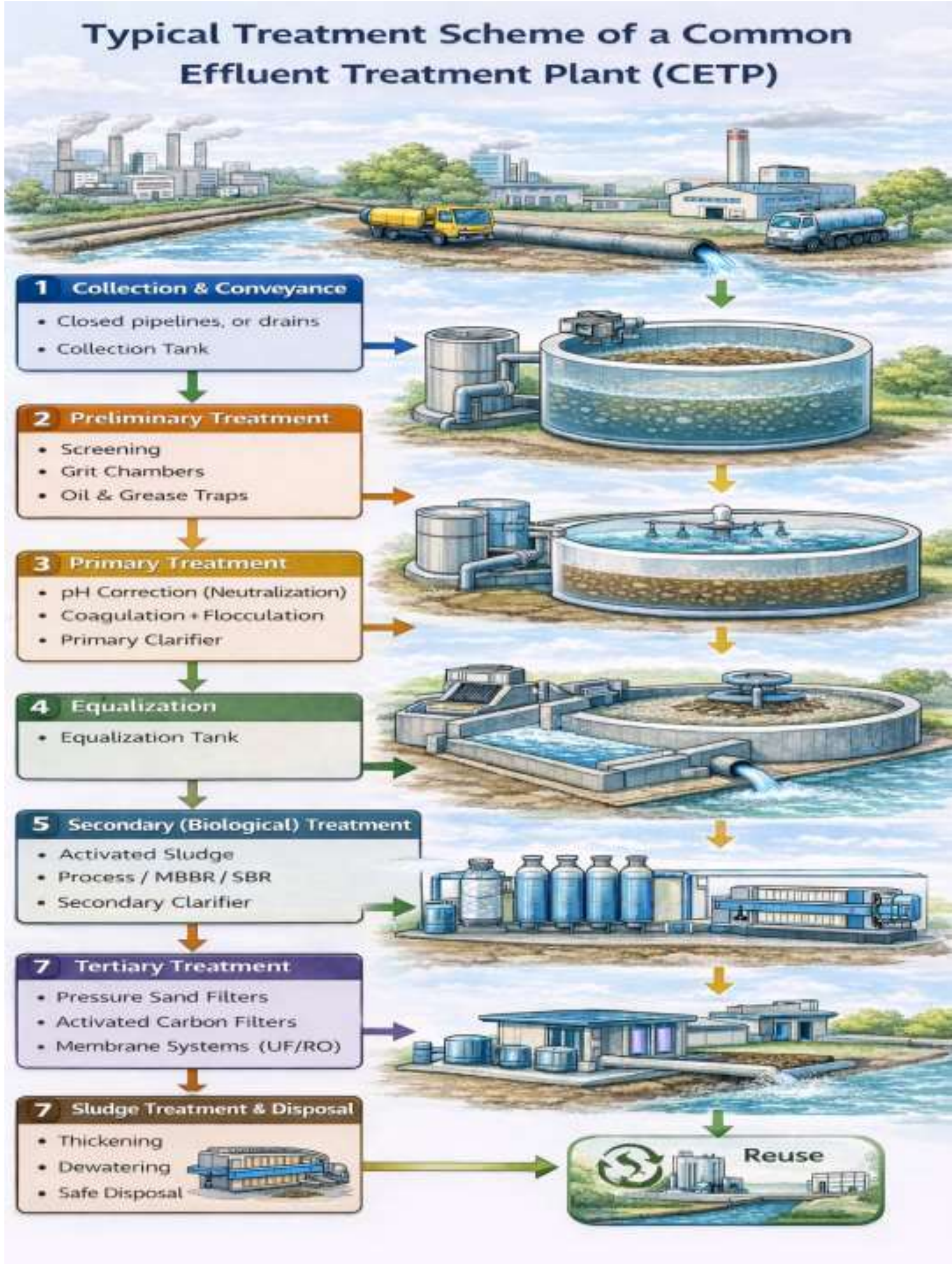
2. Preliminary Treatment

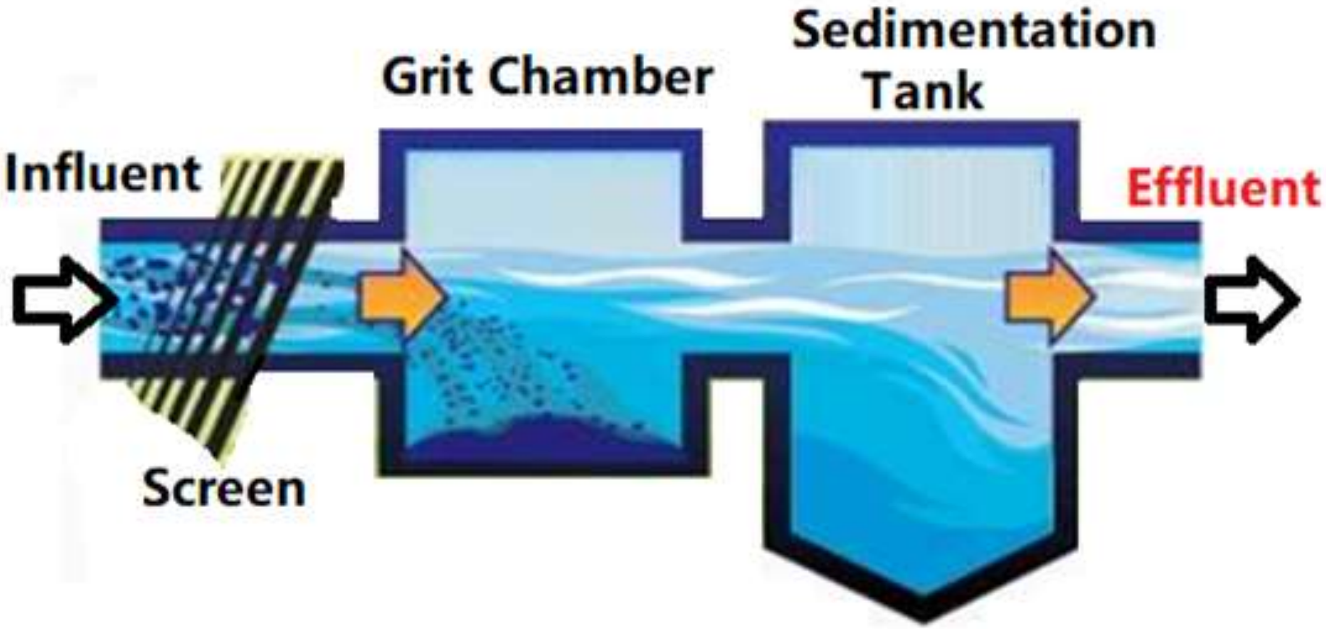
- This is the first line of protection for downstream units.

Includes:

- Screening – removes large floating materials (rags, plastics, debris)
- Grit chamber – removes sand and heavy inorganic particles
- Oil & grease trap – removes free oil and grease

Purpose: Prevent clogging, abrasion, and damage to equipment




Sedimentation
Tank
Grit Chamber
Influent
Effluent
Screen

3. Primary Treatment

Includes:

- pH correction / Neutralization (using acid or alkali)
- Coagulation and flocculation (addition of alum, ferric salts, polymers)
- Primary clarifier – settles suspended solids as sludge

Purpose:

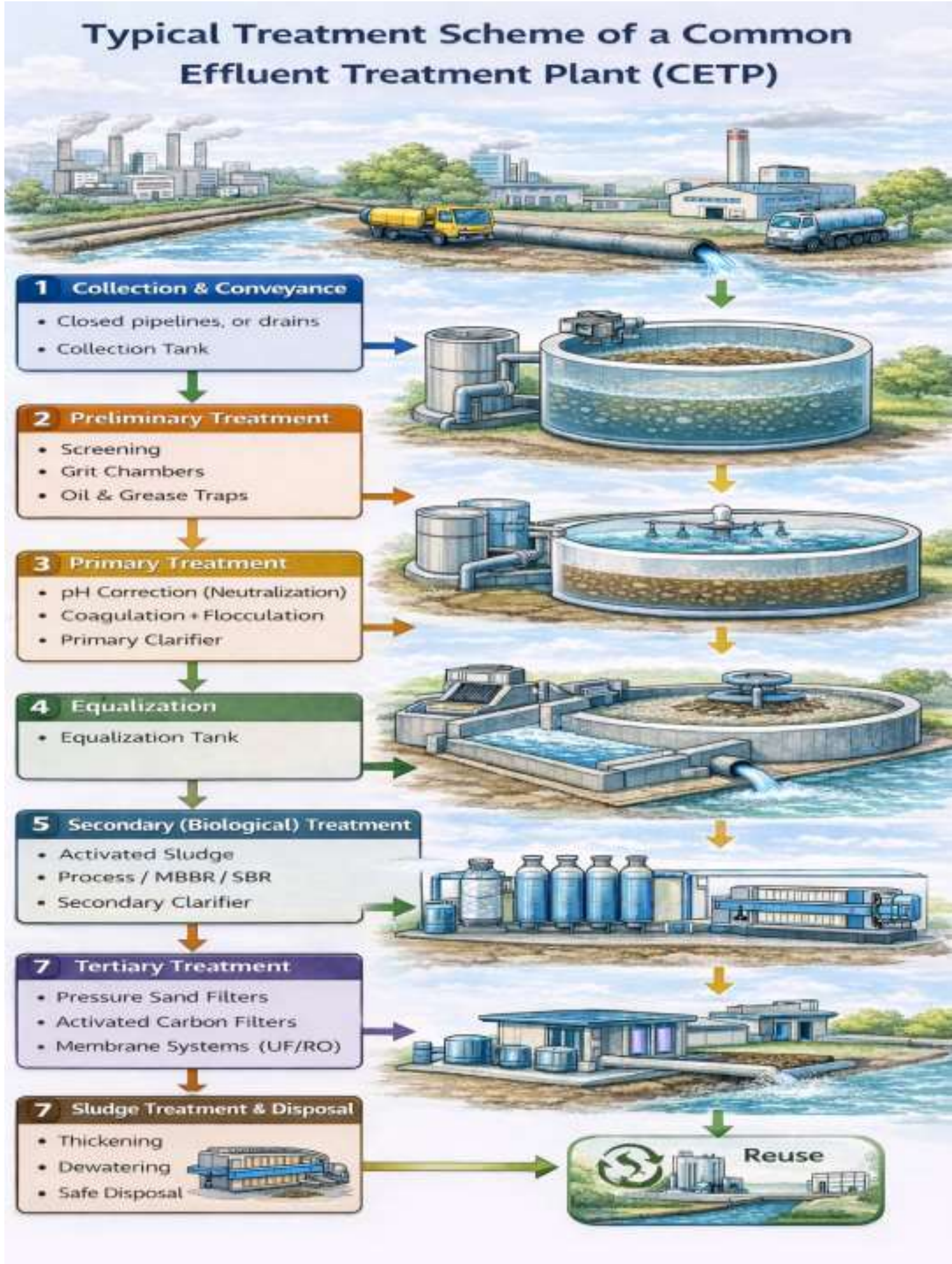
- Remove suspended solids, oil, grease
- Reduce BOD and COD partially
- Protect biological treatment units

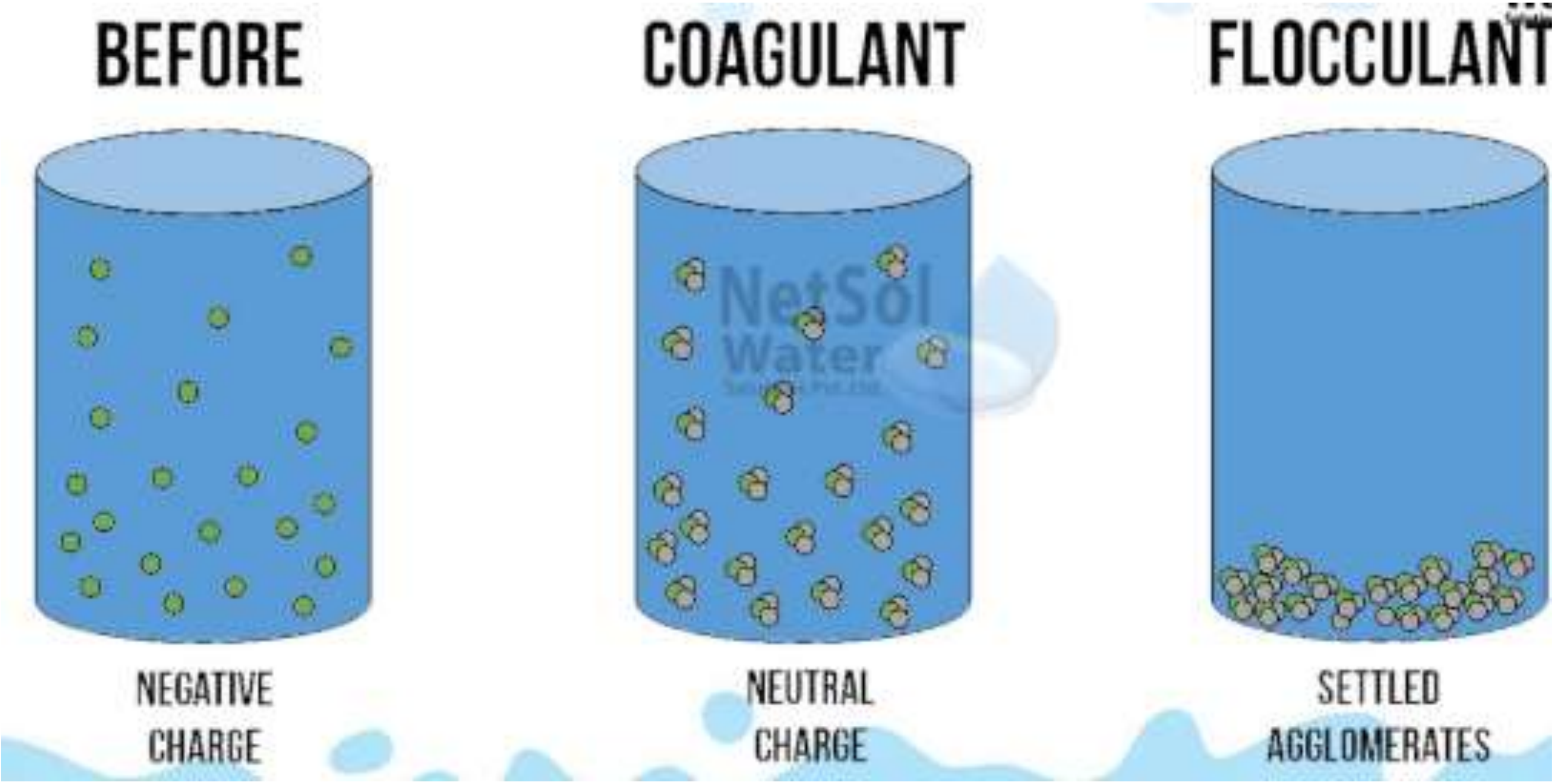
4. Equalization (with Proportioning)

- Effluent is stored in an **equalization tank**
- Continuous mixing (air or mechanical) is provided
- Sometimes **proportioning** is applied to control flow rates

Purpose:

- Reduce fluctuations in flow, pH, and pollutant load
- Provide uniform wastewater characteristics for treatment





BEFORE
COAGULANT
FLOCCULANT
NetSol Water
NEGATIVE CHARGE
NEUTRAL CHARGE
SETTLED AGGLOMERATES

5. Secondary (Biological) Treatment

- Uses microorganisms to degrade organic matter.

Common processes:

- Activated Sludge Process (ASP)
- Aerated lagoons
- Sequencing Batch Reactor (SBR)
- Followed by secondary clarifier to separate biomass

Purpose:

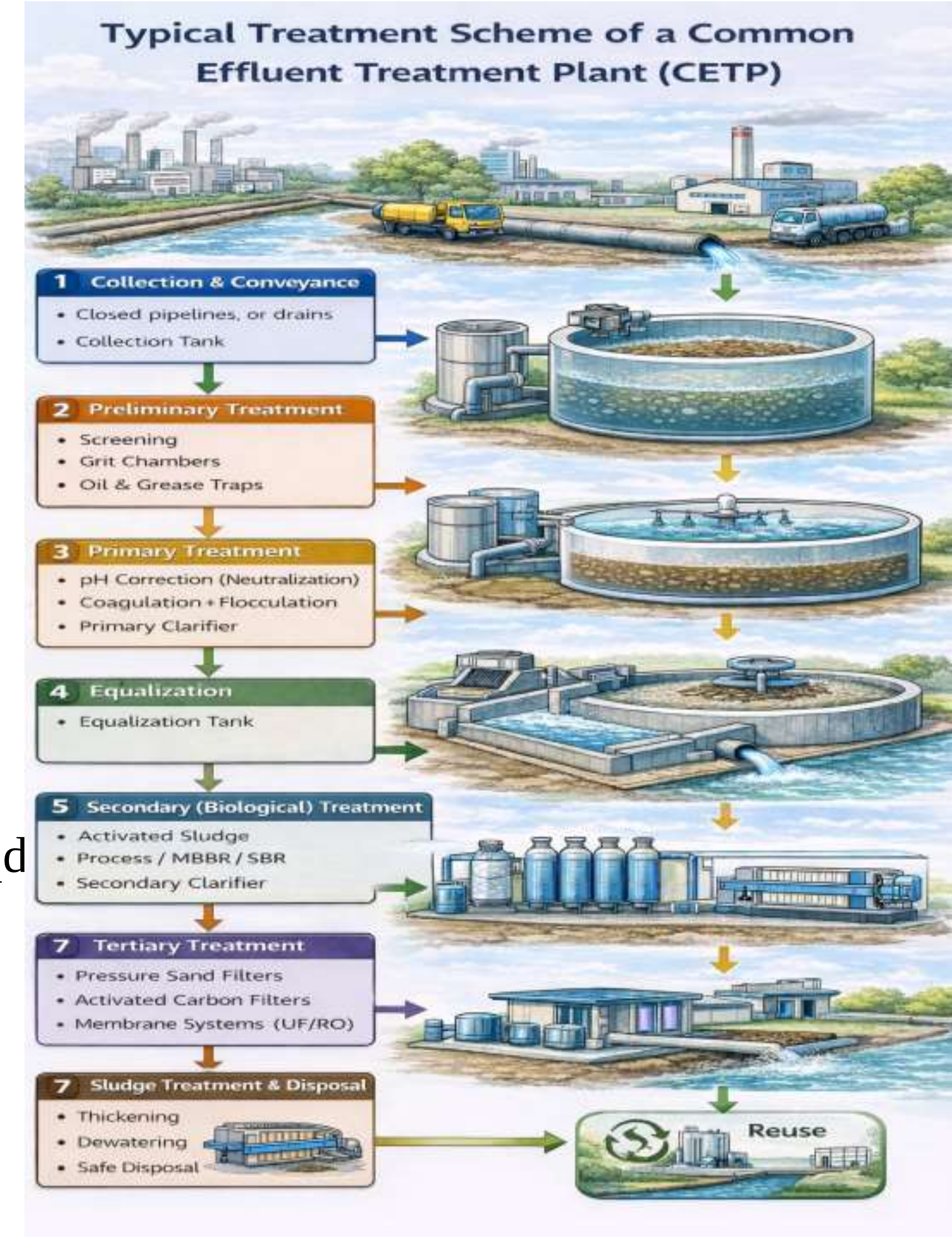
- Major reduction of BOD and COD
- Convert biodegradable organics into stable forms

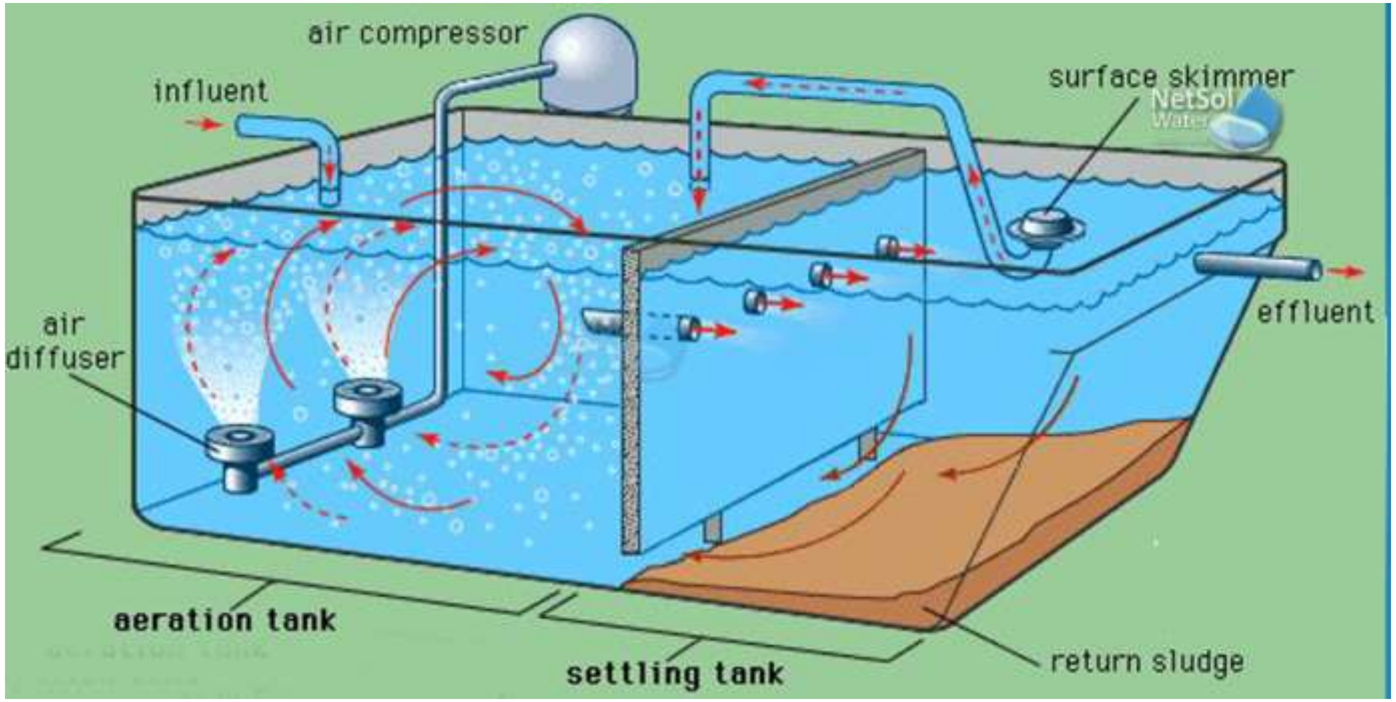
6. Tertiary / Advanced Treatment (If Required)

- Applied when stricter discharge or reuse standards are reqd

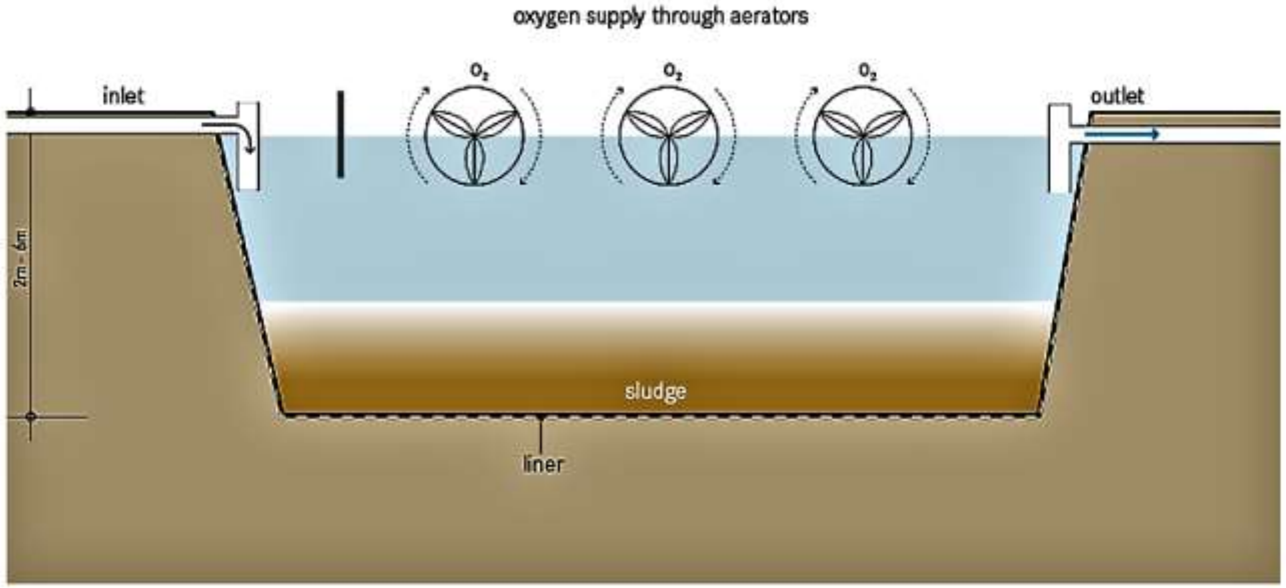
Includes:

- Pressure sand filter
- Activated carbon filter
- Membrane processes (UF / RO)
- Disinfection (chlorine/UV/ozone)





- **ASP** → Continuous flow, separate aeration & clarifier
- **SBR** → Batch process, single tank performs all steps



6. Tertiary / Advanced Treatment Remove residual solids, color, nutrients, and pathogens

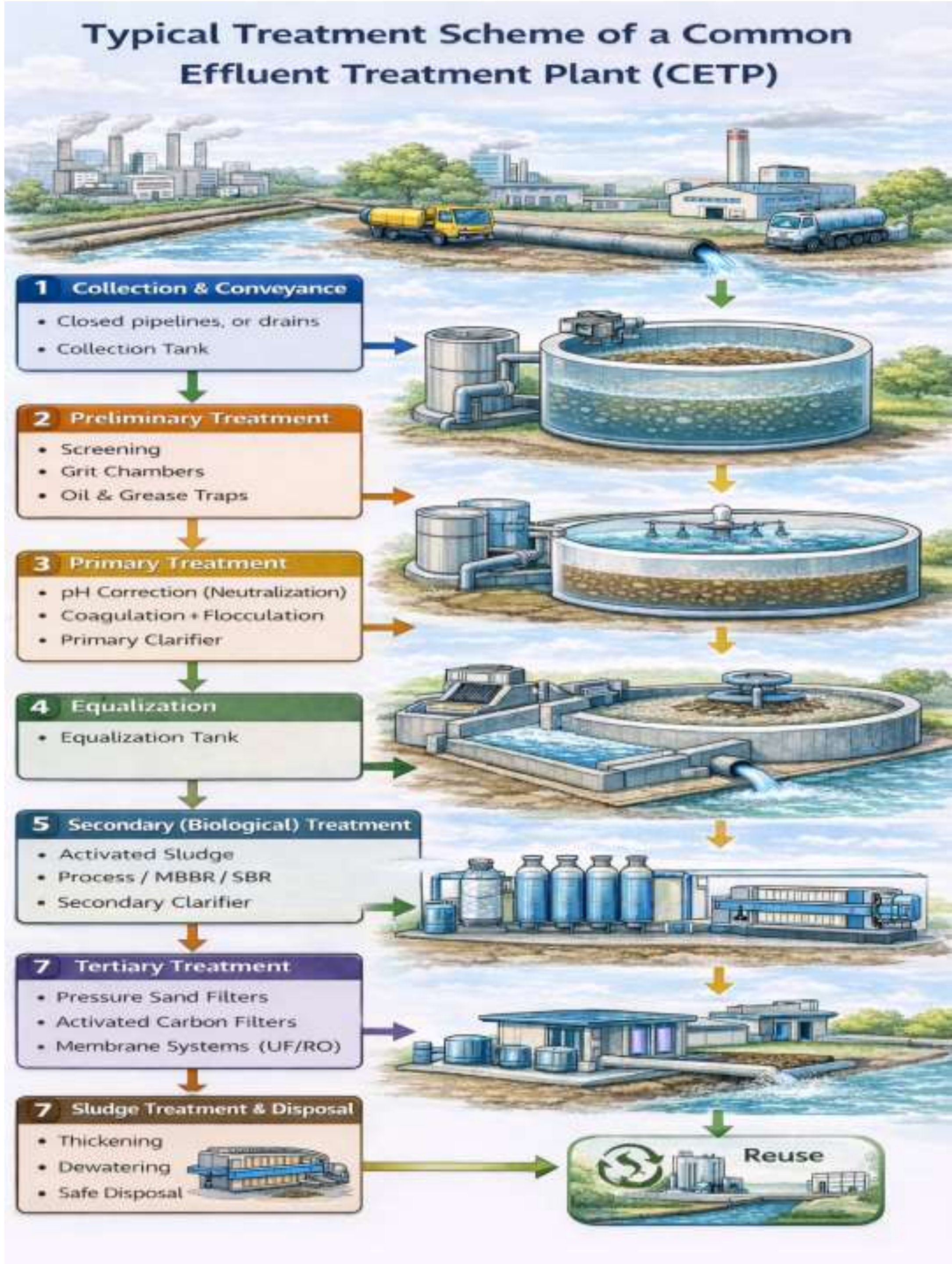
- Make water suitable for reuse or safe discharge

7. Sludge Treatment & Disposal

- Sludge generated from primary and secondary units is treated separately.

Includes:

- Sludge thickening
- Dewatering (filter press / centrifuge)
- Safe disposal or co-processing
- Purpose: Reduce volume and ensure environmentally safe disposal



Advantages

- Cost-effective.
- Professional operation and
- Compliance with pollution control norms.

Recirculation of Industrial Wastes

Definition

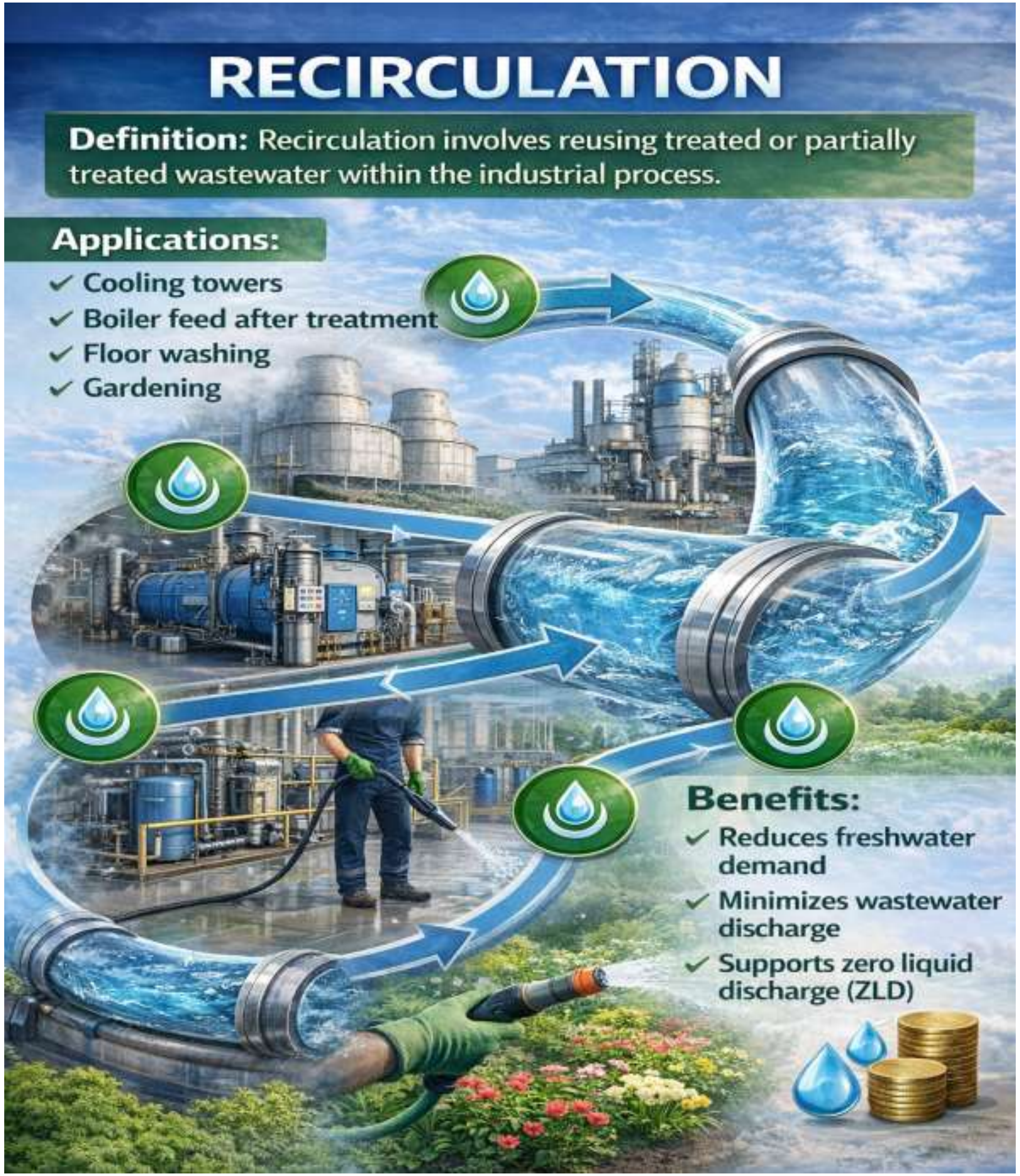
Recirculation involves reusing treated or partially treated wastewater within the same industrial process instead of discharging them as effluent

Applications

- Cooling towers.
- Boiler feed after treatment.
- Floor washing and
- Gardening.

Benefits

- Reduces freshwater demand.
- Minimizes wastewater discharge and
- Supports zero liquid discharge (ZLD).



Effluent Standards

Definition:

Effluent standards specify the **maximum permissible limits** of pollutants allowed in treated wastewater before discharge.

Typical Effluent Standards (India – CPCB)

Central Pollution Control Board

Parameter	Permissible Limit
pH	5.5 – 9.0
BOD (3 days, 27°C)	30 mg/L
COD	250 mg/L
TSS	100 mg/L
Oil & Grease	10 mg/L
Heavy metals	As specified

Importance

- Prevents environmental degradation.
- Ensures legal compliance and
- Protects public health.

Purpose of Effluent Standards

- Protect receiving water bodies.
- Prevent aquatic toxicity.
- Safeguard public health.
- Control industrial pollution.
- Ensure sustainable development.